

Socioeconomic Determinants of Cesarean Section Delivery in Bangladesh: A Survey-Weighted Statistical and Explainable Machine Learning Analysis

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Abstract:

Cesarean section delivery in Bangladesh has reached 44.65% of all births, far above the World Health Organization's recommended population threshold of 10–15%, raising concerns about over-medicalization of childbirth and inequality in maternal healthcare. This study investigates the socioeconomic and healthcare-system determinants of cesarean section delivery using nationally representative data from the Bangladesh Demographic and Health Survey 2022. The final analytical sample comprised 5,052 women. The study applies a three-stage analytical framework that combines survey-weighted generalized estimating equations for population-representative inference, six machine-learning classifiers with stratified cross-validation and Youden's J-based threshold optimization for prediction, and SHapley Additive exPlanations for global and individual-level interpretability. Bivariate analysis shows that delivery facility type is the strongest predictor, with cesarean prevalence reaching 84.3% in private facilities and remaining near zero in home deliveries (Cramér's $V = 0.763$). In the adjusted survey-weighted model, the richest wealth quintile is associated with increased odds of cesarean delivery (AOR = 1.78), four or more antenatal visits are associated with higher odds (AOR = 1.64), and fourth-or-higher-order births are associated with lower odds (AOR = 0.48). Regional variation also remained significant after adjustment, indicating important divisional differences in cesarean delivery. Across the predictive models, performance remains consistently strong, with the Stacking Ensemble achieving the best overall balance, including an F1-score of 87.06%, ROC-AUC of 0.924, and recall of 91.22%. SHAP analysis confirms delivery facility type, wealth status, and birth order as the most influential predictors and further shows that private facility delivery and intensive antenatal care contribute more strongly to cesarean risk among socioeconomically advantaged women. Overall, these findings indicate that cesarean section delivery in Bangladesh is shaped not only by obstetric factors but also by healthcare-system structure and socioeconomic inequality. The study highlights the need for stronger regulation of private maternity care, improved antenatal counselling, and equity-focused maternal health interventions.

Keywords: Cesarean Section; Survey-Weighted GEE; Explainable Machine Learning; Socioeconomic Inequality; Private Health Facilities.